BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Mixed Paper 1 and Paper 2 Retrieval

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. A £5-per-ticket program uses total = tickets + 5. What is the correction?

2. What does an IDE's run-time environment let you do?
   A) Repair physical CPU faults
   B) Rearrange HDD fragments
   C) Manage network user accounts
   D) Run and test code

3. "An IPv4 address is always six pairs of hexadecimal digits separated by colons." Correct this. Explain why.

4. An IDE finds a missing bracket. Once fixed, the program runs but gives the wrong total. Why did the IDE not prove the logic correct? What should you do next?

Today's objectives

Explain the core ideas of mixed paper 1 and paper 2 retrieval using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Synoptic retrieval
Recall previously learned Paper 1 and Paper 2 material in mixed order.

Choose the method
Decide whether a task needs an explanation, calculation, trace or evaluation.

Link systems and coding
Variables, memory and data representation affect the way programs execute.

Identify gaps
Use wrong answers to choose the next targeted practice rather than repeat every topic.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Synoptic retrieval
Recall previously learned Paper 1 and Paper 2 material in mixed order.

Choose the method
Decide whether a task needs an explanation, calculation, trace or evaluation.

Link systems and coding
Variables, memory and data representation affect the way programs execute.

Identify gaps
Use wrong answers to choose the next targeted practice rather than repeat every topic.

I Do - Worked model

A monochrome 100 x 50 bitmap uses 1 bit/pixel. Calculate its uncompressed file size in bytes.

- Compute pixels: 100 x 50=5000.
- Multiply by one bit per pixel: 5000 bits.
- Convert 5000 / 8=625 bytes.
- State assumptions: ignoring metadata and compression.

We Do - Guided practice

1. A mono sound clip lasts 5 seconds at 8000 samples/s and 8 bits/sample. Find bytes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the MOD calculation that finds leftover bits.

```
bits = 203
leftover = ...
print(leftover)
```

You Do - Independent exam questions

Q2. Calculate the denary value of unsigned binary 10101010. [3 marks]

Q3. Explain why binary search requires a sorted list. [3 marks]

Q4. Explain one advantage and one limitation of virtual memory. [4 marks]

Exit Ticket - True or False?

Image file size in bytes equals width x height x bit depth without any conversion.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Use total = tickets * 5; addition does not calculate £5 for each ticket.

2. D - Run and test code

3. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

4. The IDE found a syntax error, not a logic error. Trace the calculation. Test inputs with known totals and compare results.

I Do - Explain the model

Compute pixels: 100 x 50=5000.

Multiply by one bit per pixel: 5000 bits.

Convert 5000 / 8=625 bytes.

State assumptions: ignoring metadata and compression.

We Do - Model answers

1. 40,000 bytes.

Code First - Answer

```
bits = 203
leftover = bits MOD 8
print(leftover)
```

Expected output: 3

You Do - Self-mark

Q2: 128+32+8+2 = 170.

Q3: Midpoint comparisons only allow safe elimination of half the list when values are ordered.

Q4: Lets the OS accommodate memory demand beyond physical RAM; uses slower secondary storage, so paging can reduce performance.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.